

Chapter 1

BCR (Benefit-Cost Ratio) Cheat Sheet

Quick reference guide for all BCR perspectives calculated by CTCC.

Important Note on Revenue

Revenue (rate-based revenue requirement) is a **transfer** from ratepayers to the utility/TSP, not a net social benefit. System/societal BCRs exclude revenue from total benefits. Revenue appears only in stakeholder BCRs where it's a benefit to utilities and a cost to ratepayers.

Cost categories

Capital = build + ROW capital (acquisition + holding) + environmental mitigation.
Operational = O&M + insurance + ROW rent. **Energy/Emissions** = line losses + emissions + residual exceedance. Residual exceedance is a system/societal cost only. Rate base = AFUDC capital (build + row capital + env) at COD. Revenue is computed in real terms (Option A: real stream, real WACC).

1.1 BCR Metrics (order matches Benefit-Cost Ratio Metrics tables)

Order below matches the per-scenario and all-scenarios BCR tables in `results_tables.tex` (updated by `papers/update_results_tables.py`).

1.1.1 `bcr_system`

Equation: Total Benefits (haircut) / Total Costs

Perspective: Full societal/system-wide analysis

Includes:

- **Benefits:** Congestion reduction + Curtailment reduction (with conservative haircuts; excludes revenue transfer)

- **Costs:** Capital + Operational (O&M + insurance only) + Energy/Emissions (line losses + emissions + residual exceedance) + Risk + Delay (everything)

Use Case: Complete societal cost-benefit analysis including all externalities and risks.

1.1.2 `bcr_utility`

Equation: Revenue / (Capital Costs + Delay Costs + Operational Costs). Benefits: Revenue only. Costs: Capital + Delay + Operational (O&M + insurance only). Excludes: Line losses, Emissions, Residual exceedance, Risk.

Key Insight: Utilities don't receive congestion/curtailment benefits directly; they only get rate base recovery.

1.1.3 `bcr_ratepayer`

Equation: (Congestion Benefits + Curtailment Benefits) / (Revenue + Line Loss Costs). Benefits: Congestion + Curtailment. Costs: Revenue + Line losses. Excludes: Capital/O&M as separate line items (recovered via Revenue).

Key Insight: Ratepayers receive congestion/curtailment relief but pay revenue/rate base and line losses through rates.

1.1.4 `bcr_capital`

Equation: Total Benefits (haircut) / Capital Costs. Costs: Build + ROW + environmental only.

1.1.5 `bcr_capital_and_delay`

Equation: Total Benefits (haircut) / (Capital Costs + Delay Costs). Costs: Capital + Delay (construction delay + congestion delay + curtailment delay).

1.1.6 `bcr_primary`

Equation: Primary Benefits / Primary Costs (customizable via flags). Benefits: Congestion (optional) + Curtailment (optional). Costs: Capital + Delay (always) + O&M, Insurance, Wildfire, Outage, Line Losses, Emissions (optional). If no flags are set, equals `bcr_system`.

1.1.7 `bcr_excluding_wildfire_risk_and_outage_risk`

Equation: Total Benefits (haircut) / (Total Costs - Wildfire Risk - Outage Risk)

Perspective: Societal perspective excluding both wildfire and outage risk costs.

Includes: Benefits: Congestion + Curtailment (excludes revenue). Costs: Capital + Operational + Energy/Emissions + Delay (excludes wildfire + wildfire liability + outage).

Use Case: Analysis when both wildfire and outage risk costs may be insured, managed separately, or contextually inappropriate.

1.1.8 `bcr_excluding_emissions`

Equation: Total Benefits (haircut) / (Total Costs - Emissions Costs). Costs: Capital + Operational + Risk + Delay + Line Losses (excludes emissions only).

1.1.9 `bcr_excluding_linelosses`

Equation: Total Benefits (haircut) / (Total Costs - Line Loss Costs). Costs: Capital + Operational + Risk + Delay + Emissions (excludes line losses only).

1.1.10 `bcr_excluding_emissions_and_linelosses`

Equation: Total Benefits (haircut) / (Total Costs - Emissions Costs - Line Loss Costs). Costs: Capital + Operational + Risk + Delay (excludes emissions + line losses).

1.1.11 `bcr_excluding_emissions_and_wildfire_risk_and_outage_risk`

Equation: Total Benefits / (Total Costs - Emissions - Wildfire Risk - Outage Risk). Costs: Capital + Operational + Delay + Line Losses.

1.1.12 `bcr_excluding_linelosses_and_wildfire_risk_and_outage_risk`

Equation: Total Benefits / (Total Costs - Line Losses - Wildfire Risk - Outage Risk). Costs: Capital + Operational + Delay + Emissions.

1.1.13 `bcr_excluding_emissions_and_linelosses_and_wildfire_risk_and_outage_risk`

Equation: Total Benefits / (Total Costs - Emissions - Line Losses - Wildfire Risk - Outage Risk). Costs: Capital + Operational + Delay only.

1.1.14 `bcr_excluding_wildfire_risk`

Equation: Total Benefits (haircut) / (Total Costs - Wildfire Risk Costs). Costs: Capital + Operational + Energy/Emissions + Delay + Outage Risk (excludes wildfire + wildfire liability).

1.1.15 `bcr_excluding_emissions_and_wildfire_risk`

Equation: Total Benefits / (Total Costs - Emissions - Wildfire Risk). Costs: Capital + Operational + Delay + Line Losses + Outage Risk.

1.1.16 `bcr_excluding_linelosses_and_wildfire_risk`

Equation: Total Benefits / (Total Costs - Line Loss Costs - Wildfire Risk Costs). Costs: Capital + Operational + Delay + Emissions + Outage Risk.

1.1.17 `bcr_excluding_emissions_and_linelosses_and_wildfire_risk`

Equation: Total Benefits / (Total Costs - Emissions - Line Losses - Wildfire Risk). Costs: Capital + Operational + Delay + Outage Risk only.

1.1.18 `bcr_excluding_outage_risk`

Equation: Total Benefits (haircut) / (Total Costs - Outage Risk Costs). Costs: Capital + Operational + Energy/Emissions + Delay + Wildfire Risk (excludes outage).

1.1.19 `bcr_excluding_emissions_and_outage_risk`

Equation: Total Benefits / (Total Costs - Emissions - Outage Risk). Costs: Capital + Operational + Delay + Line Losses + Wildfire Risk.

1.1.20 `bcr_excluding_linelosses_and_outage_risk`

Equation: Total Benefits / (Total Costs - Line Loss Costs - Outage Risk Costs). Costs: Capital + Operational + Delay + Emissions + Wildfire Risk.

1.1.21 `bcr_excluding_emissions_and_linelosses_and_outage_risk`

Equation: Total Benefits / (Total Costs - Emissions - Line Losses - Outage Risk). Costs: Capital + Operational + Delay + Wildfire Risk only.

1.2 Net Benefits

All BCRs have corresponding net benefit calculations (Present Value unless noted).

1.2.1 System/Societal Net Benefits

- `net_benefit_pv` = Total Benefits (haircut) - Total Costs
- `net_benefit_nominal` = Total Benefits (nominal) - Total Costs (nominal), undiscounted
- `net_benefit_primary_pv` = Primary Benefits - Primary Costs (customizable)
- `net_benefit_excluding_wildfire_risk_and_outage_risk_pv` = Total Benefits - (Total Costs - Wildfire Risk - Outage Risk)
- `net_benefit_excluding_wildfire_risk_pv`, `net_benefit_excluding_outage_risk_pv`
- `net_benefit_excluding_emissions_and_linelosses_pv`
- `net_benefit_excluding_emissions_and_linelosses_and_wildfire_risk_and_outage_risk_pv`
- `net_benefit_capital_only_pv`, `net_benefit_capital_and_delay_pv`

1.2.2 Stakeholder Net Benefits

- $\text{net_benefit_utility_pv} = \text{Revenue} - (\text{Capital} + \text{Delay} + \text{Operational Costs})$
- $\text{net_benefit_ratepayer_pv} = (\text{Congestion} + \text{Curtailment Benefits}) - (\text{Revenue} + \text{Line Loss Costs})$

1.2.3 Partial Exclusion Net Benefits (16 Combinations)

Systematic exploration of all combinations excluding emissions, line losses, wildfire risk, and outage risk. All use societal perspective (benefits: congestion + curtailment; excludes revenue transfer). Positive = benefits exceed costs; Negative = costs exceed benefits.

WF	Out	Em	LL	Net Benefit Variable	Equation
N	N	N	N	<code>net.benefit.pv</code>	Benefits - Total Costs
Y	N	N	N	<code>...excl. wildfire_risk.pv</code>	Benefits - (Costs - WF)
N	Y	N	N	<code>...excl. outage_risk.pv</code>	Benefits - (Costs - Outage)
Y	Y	N	N	<code>...excl. wildfire_and_outage.pv</code>	Benefits - (Costs - WF - Out)
N	N	Y	N	<code>...excl. emissions.pv</code>	Benefits - (Costs - Em); keeps LL
Y	N	Y	N	<code>...excl. emissions_and_wildfire.pv</code>	Benefits - (Costs - Em - WF)
N	Y	Y	N	<code>...excl. emissions_and_outage.pv</code>	Benefits - (Costs - Em - Out)
Y	Y	Y	N	<code>...excl. em_wf_outage.pv</code>	Benefits - (Costs - Em - WF - Out)
N	N	N	Y	<code>...excl. linelosses.pv</code>	Benefits - (Costs - LL); keeps Em
Y	N	N	Y	<code>...excl. LL_wildfire.pv</code>	Benefits - (Costs - LL - WF)
N	Y	N	Y	<code>...excl. LL_outage.pv</code>	Benefits - (Costs - LL - Out)
Y	Y	N	Y	<code>...excl. LL_wf_outage.pv</code>	Benefits - (Costs - LL - WF - Out)
N	N	Y	Y	<code>...excl. emissions_and_linelosses.pv</code>	Benefits - (Costs - Em - LL)
Y	N	Y	Y	<code>...excl. em_LL_wildfire.pv</code>	Benefits - (Costs - Em - LL - WF)
N	Y	Y	Y	<code>...excl. em_LL_outage.pv</code>	Benefits - (Costs - Em - LL - Out)
Y	Y	Y	Y	<code>...excl. em_LL_wf_outage.pv</code>	Benefits - (Costs - Em - LL - WF - Out)

Column key: WF = wildfire risk, Out = outage risk, Em = emissions, LL = line losses. “...” = `net_benefit_excluding_..`

1.3 Quick Decision Guide

Question	Use This BCR
Should a utility/TSP build this?	<code>bcr_utility</code>
Do ratepayers benefit?	<code>bcr_ratepayer</code>
Is it good for society overall?	<code>bcr_system</code> or <code>bcr_excluding_wildfire_risk_and_outage_risk</code>
Is the capital investment attractive?	<code>bcr_capital</code> or <code>bcr_capital_and_delay</code>
Ignore wildfire risk (keep outage)?	<code>bcr_excluding_wildfire_risk</code>
Ignore outage risk (keep wildfire)?	<code>bcr_excluding_outage_risk</code>
Ignore both wildfire and outage risk?	<code>bcr_excluding_wildfire_risk_and_outage_risk</code>
Ignore emissions only?	<code>bcr_excluding_emissions</code>
Ignore line losses only?	<code>bcr_excluding_line losses</code>
Ignore both emissions and line losses?	<code>bcr_excluding_emissions_and_line losses</code>
Ignore emissions and both risks?	<code>bcr_excluding_emissions_and_wildfire_risk_and_outage_risk</code>
Ignore line losses and both risks?	<code>bcr_excluding_line losses_and_wildfire_risk_and_outage_risk</code>
Ignore emissions and wildfire risk (keep outage)?	<code>bcr_excluding_emissions_and_wildfire_risk</code>
Ignore emissions and outage risk (keep wildfire)?	<code>bcr_excluding_emissions_and_outage_risk</code>
Core project economics only (no externalities, no risks)?	<code>bcr_excluding_emissions_and_line losses_and_wildfire_risk_and_outage_risk</code>

1.4 Important Notes

1. **Haircut Benefits:** Most BCRs use conservative “haircut” benefits (saturation factors applied to congestion/curtailment).
2. **Present Value:** All values are in present value (PV) terms, discounted to base year.
3. **Context:** Wildfire costs assume California-style catastrophic events. For lower wildfire risk regions, `bcr_excluding_wildfire_risk` may be more appropriate. Outage risk is more universal. Line losses are socialized through rates; emissions are externalities.
4. **Separating Wildfire and Outage Risk:** Use `bcr_excluding_wildfire_risk` for Virginia-style scenarios; `bcr_excluding_wildfire_risk_and_outage_risk` when both should be excluded; `bcr_excluding_outage_risk` when only outage risk should be excluded.

1.5 Example: Scenario 1 (Rural Overhead AC 657MW)

- `bcr_utility` = 0.36 → Utility loses money.

- `bcr_ratepayer` = 0.22 $\rightarrow$ Ratepayers lose money.
- `bcr_system` = 0.02 $\rightarrow$ Very poor societal return (wildfire risk dominates).
- `bcr_excluding_wildfire_risk_and_outage_risk` = 0.24 $\rightarrow$ Better but still negative.

The separation of wildfire and outage risk allows for more nuanced regional analysis.